DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

WO-500-04 — Replace VALVE, BALL, FLOATING 1" P150 (1"-P150-HS-5011/A3)

Area	A500 Pipe rack	**Task**	VR Valve replacement in kind
Priority	3 (reliability)	**Issued by**	Facility owner (demonstration)
Primary asset	**rack-bent:720** — Pipe rack bent at E 60'-0" — E 60'-0" N 0.000" EL 122'-0"		
Equipment	—	**Lines**	1"-P150-HS-5011
Isometrics	ISO-1"-P150-HS-5011 (generated); owner's drawings ISO-HS-5011		
Permits	General work permit		

Reason

Hose-station take-off valve at the water header seized.

Scope of work

1. Isolate per the isolation list, drain and vent 1"-P150-HS-5011 between the isolations.
2. Unbolt and remove VALVE, BALL, FLOATING 1" P150 (1"-P150-HS-5011/A3) at E 62'-6" N -4'-0" EL 123'-7-1/2".
3. Inspect both flange faces (SW-NDE-5); dress or report damage.
4. Install a new VALVE, BALL, FLOATING 1" P150 of the same size, class and face-to-face (BALL1-P150), with new gaskets and studs, stem orientation as removed.
5. Tighten the studs to the stated preload in three passes, cross pattern (SW-WO-5).
6. Remove blinds, restore the line-up, leak test the two joints on start-up (SW-TEST-2).

Affected items: 1"-P150-HS-5011/A3, 1"-P150-HS-5011/A2, 1"-P150-HS-5011/A5

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-5, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-HS-5011	1"-P150-HS-5011	1"-P150-HS-5011	1	agrees with the model

ISO-HS-5011 (PDF) — the generated isometrics follow the ITP.

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
rack-bent:720	structure	Pipe rack bent at E 60'-0"		E 60'-0" N 0.000" EL 122'-0"
1"-P150-HS-5011/A3	item	VALVE, BALL, FLOATING 1" P150	1"-P150-HS-5011	E 62'-6" N -4'-0" EL 123'-7-1/2"
1"-P150-HS-5011/A2	item	GASKET 1" P150 COMPRESSED SHEET	1"-P150-HS-5011	E 62'-6" N -4'-0" EL 123'-3/4"
1"-P150-HS-5011/A5	item	GASKET 1" P150 COMPRESSED SHEET	1"-P150-HS-5011	E 62'-6" N -4'-0" EL 124'-2-1/4"
3"-P150-W-2008/T:1"-P150-HS-5011	item	TEE REDUCING 3" × 1" P150	3"-P150-W-2008	E 62'-6" N -4'-0" EL 122'-6"
1"-P150-HS-5011/B3	item	VALVE, BALL, FLOATING 1" P150	1"-P150-HS-5011	E 62'-6" N 11'-0" EL 104'-2-1/8"
1"-P150-HS-5011/B6	item	FLANGE WELD-NECK 1" P150 RAISED FACE	1"-P150-HS-5011	E 62'-6" N 11'-0" EL 104'-11-1/16"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	equipment	3"-P150-W-2008/T:1"-P150-HS-5011	3"-P150-W-2008	No valve on this side within the line: isolate or depressure the parent line 3"-P150-W-2008 at its tee under the shutdown isolation			
2	valve	1"-P150-HS-5011/B3	1"-P150-HS-5011	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-HS-5011/B3)	E 62'-6" N 11'-0" EL 104'-2-1/8"		
3	blind	1"-P150-HS-5011/B6	1"-P150-HS-5011	Fit a line blind between 1"-P150-HS-5011/B3 and FLANGE WELD-NECK 1" P150 RAISED FACE (1"-P150-HS-5011/B6), work side	E 62'-6" N 11'-0" EL 104'-8-15/16"		

Parts

Description	Qty	Replaces
VALVE, BALL, FLOATING 1" P150	1	1"-P150-HS-5011/A3

Description	Qty	Replaces
GASKET 1" P150 COMPRESSED SHEET	1	1"-P150-HS-5011/A2
STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150	1	1"-P150-HS-5011/A4
GASKET 1" P150 COMPRESSED SHEET	1	1"-P150-HS-5011/A5
STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150	1	1"-P150-HS-5011/A7

Inspections

Id	Inspection	Criteria	Basis
WO-500-04-I1	Flange faces of both joints before the new gaskets go in	No radial scratch across the sealing face (SW-NDE-5)	SW-NDE-5

Bolt-up (SW-WO-5)

Joint	Studs	Preload per stud (lbf)	Passes	Done ✓
1"-P150-HS-5011/A2	4 × 0.5" studs (1"-P150-HS-5011/A4)	8,836	3	
1"-P150-HS-5011/A5	4 × 0.5" studs (1"-P150-HS-5011/A7)	8,836	3	

Test — Service leak test

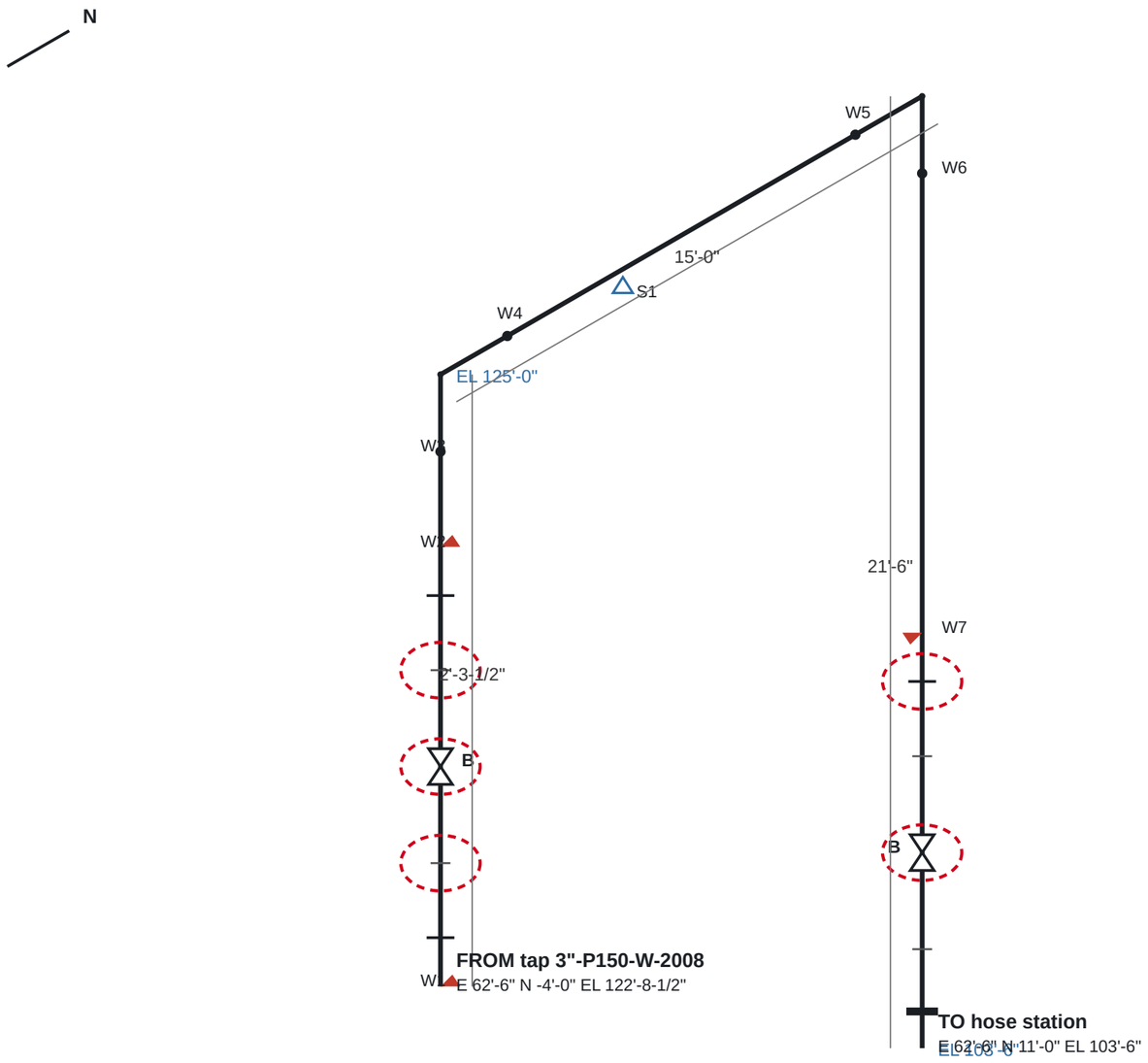
Description	Both remade joints checked at operating pressure on start-up
Basis	SW-TEST-2

Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Valve 1"-P150-HS-5011/A3 removed; flange faces inspected	SW-NDE-5	No radial scratch across the sealing face	Flange face inspection record	W	W	—				
3	New valve verified: type, size, class, face-to-face and tag	BALL1-P150	Matches the parts list and the line class	Material receiving record	R	R	—				
4	New gaskets fitted, studs tightened to preload in three passes	SW-WO-5	Preload per stud as stated; three passes, cross pattern	Bolt-up record per joint	W	W	—				
5	Service leak test of the remade joints on start-up	SW-TEST-2	No leak at either joint	Leak test record	W	W	—				
6	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
7	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

ISOMETRIC — 1"-P150-HS-5011 — Replace VALVE, BALL, FLOATING 1" P150 (1"-P150-HS-5011/A3)



BILL OF MATERIALS

1	FLANGE WELD-NECK 1" P150 RAISED FACE	3
2	GASKET 1" P150 COMPRESSED SHEET	4
3	VALVE, BALL, FLOATING 1" P150	2
4	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150	4
5	PIPE 1" P150 BEVELLED ENDS	34'-9-11/16"
6	ELBOW 90° R = 5/3 OD 1" P150	2
7	FLANGE BLIND 1" P150 RAISED FACE	1

WELDS 7 (3 FIELD) · SUPPORTS 1 · SPOOLS 4

W1F W2F W3 W4 W5 W6 W7F

1"-P150-HS-5011

CLASS W150 — Water (cooling and process)
SERVICE HS · 1" · P150 · DESIGN 150 psi
tap 3"-P150-W-2008 → hose station
PIPE 34'-9-11/16" · WEIGHT 120 lb

DWG ISO-1"-P150-HS-5011 REV 0 WO-500-04

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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